A Fault-Tolerant Decentralized Coordination Framework for UAV Swarms using Relative Suitability Elective Consensus and SITL Validation

A project report submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

by

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Under the Guidance of

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Professor & Chairman, Board of Studies

Dept. of Computer Science and Systems Engineering

Department of Computer Science and Systems Engineering

Andhra University College of Engineering (A)

Andhra University, Visakhapatnam – 530003, India

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CERTIFICATE

This is to certify that the project report entitled **“A Fault-Tolerant Decentralized Coordination Framework for UAV Swarms using Relative Suitability Elective Consensus and SITL Validation”** is a bonafide project work being carried out by **Mohamed Hijazy Shazin Hassan (322106410064)**, **Mohsen Hasan Mohammed Ba Haj (322106410065)**, **Hassan Said Abdi (322106410058) and Gurumalla Sai Nikhil (322106410019)** in partial fulfillment of the requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** from the Department of Computer Science & Systems Engineering, Andhra University College of Engineering (A), Andhra University, Visakhapatnam.

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DECLARATION

We hereby declare that the project report entitled **"A Fault-Tolerant Decentralized Coordination Framework for UAV Swarms using Relative Suitability Elective Consensus and SITL Validation"** is an original work done at the Department of Computer Science and Systems Engineering, Andhra University College of Engineering (A), Andhra University, Visakhapatnam, under the guidance of Prof. Kuda Nageswara Rao and submitted in partial fulfillment of the requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** during December 2025 – May 2026. The same work either in part or whole has not been submitted to any other institution.

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ABSTRACT

The deployment of multi-agent Unmanned Aerial Vehicle (UAV) swarms has revolutionized autonomous operations in complex, dynamic environments, ranging from search-and-rescue missions to tactical reconnaissance. However, the reliability and scalability of these robotic networks are frequently bottlenecked by their reliance on centralized control architectures. This traditional topology inherently introduces a critical single point of failure; any hardware degradation, communication blackout, or power depletion at the central command node instantly jeopardizes the structural integrity and mission continuity of the entire swarm.

To resolve these critical vulnerabilities, this thesis presents **A Fault-Tolerant Decentralized Coordination Framework for UAV Swarms using Relative Suitability Elective Consensus and SITL Validation**, a highly autonomous, fault-tolerant simulation framework engineered to eliminate the need for centralized command oversight. The system is architected using the Robot Operating System 2 (ROS 2 Humble Hawksbill) middleware to handle high-level decentralized communication via Data Distribution Service (DDS) topics. To achieve high-fidelity aerodynamic realism, the architecture integrates the industry-standard PX4-Autopilot flight stack and MAVSDK C++ API for offboard control, operating within a simulated Gazebo Classic 11 3D physics environment. The entire Software-in-the-Loop (SITL) ecosystem is deployed within a host system.

The fundamental contribution of this research is a continuous, fully decentralized leadership election algorithm. Rather than relying on pre-programmed hierarchies, individual agents autonomously evaluate their suitability to coordinate the swarm through a periodically computed mathematical construct defined as the Relative Suitability Value (RSV). During synchronized computational cycles, every identical drone within the network independently calculates its RSV by evaluating real-time local telemetry. This multi-variable equation mathematically weighs three critical parameters: residual battery energy (E_{res}), localized communication neighbour density (N_{deg}), and physical Euclidean distance to the swarms calculated centroid (D_{ctr}).

The agent achieving the highest RSV seamlessly transitions into the role of Node Leader, assuming responsibility for generating mission trajectories via MAVSDK velocity setpoints, while simultaneously broadcasting a continuous 1 Hz status heartbeat over the local network. To guarantee operational safety, the architecture features a rigorous, state-driven failover protocol. Follower drones actively monitor the ROS 2 DDS network; if the active leader experiences failure and its heartbeat drop out beyond a strict 3.0-second threshold ($H_{timeout}$), the agent possessing the second-highest suitability score instantly executes an autonomous takeover. By distributing computational logic uniformly across the network, the Decentralized Adaptive Swarm Control System demonstrates exceptional structural resilience, optimized computational load balancing, and rapid, seamless failure recovery.

Keywords: Decentralized Swarm Intelligence, UAV Coordination, ROS 2 Humble, PX4-Autopilot, Relative Suitability Value (RSV), Leader Election Algorithm, Fault-Tolerant Systems, MAVSDK, Software-in-the-Loop (SITL), Peer-to-Peer Communication, Autonomous Failover, Gazebo Simulation.